

SACHIN THOMAS

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EDUCATION

University of Massachusetts, Amherst

◦ Master of Science in Computer Science - GPA: 4/4

- Courses: Applied ML in Child Rescue, Software Engineering, Computer and Network Security, Machine Learning, Algorithms for Data Science, Secure Distributed Systems, Database Design and Implementation, Responsible AI

Massachusetts, USA

Sep 2023 - May 2025

National Institute of Technology, Calicut

◦ Bachelor of Technology in Computer Science and Engineering - GPA: 8.55/10

Kerala, India

Aug 2019 - May 2023

SKILLS

Technical skills: C++, C, Python, Object Oriented Programming, Data Structures and Algorithms, MySQL, PostgreSQL, React, Flask, Javascript, Typescript, Internet Protocols (HTTPS, TCP/IP, SMTP, FTP), Machine Learning, Pytest

EXPERIENCE

Graduate Research Assistant, UMass Amherst

February 2024 - Present

- Conducted an in-depth analysis of IoT device companion apps to assess potential security and privacy risks.
- Investigated permission requests and network communication patterns to identify unauthorized data access, excessive permissions, and potential vulnerabilities.

Teaching Assistant, UMass Amherst

September 2024 - December 2024

- Assisted in teaching core software engineering concepts, including software development lifecycle, design patterns, and best coding practices.
- Conducted office hours to help students with programming assignments, debugging, and understanding course materials.
- Graded assignments and projects, providing detailed feedback to enhance student learning.

iXp Intern (Software Development) - SAP Labs

Bangalore, India

May 2022 - July 2022

◦ Situation Handling

- Worked on integrating various internal SAP products to implement a notification feature that raises alerts via a risk score metric to improve response time by 30%.

◦ Public Sector Management - SAC Integration

- Developed ODATA APIs for automation of budget planning that reduced manual effort involved in the testing process by 60% while improving productivity and ensuring the consistency and accuracy of the API implementations.

Software Development Intern - Parametricos

Cyprus (Distant Internship)

March 2021 - July 2021

◦ Studio 3DX

- Built a fast parsing solution for IFC files in C++ and TypeScript to reduce render times by 99% in collaboration with members of the Software Development Team.

◦ Open Source Plugin - ifcGH

- Worked on ifcGH plugin for Grasshopper Rhino in C# (open-source) to expand its coverage of compatible IfcEntity types to 90%.

PROJECTS

Video Summarizer (Student Researcher - UMass Amherst Rescue Lab/Mass Law Enforcement)

- Developed an automated video summarization pipeline using FFmpeg, Google's Gemma, and OpenAI's Whisper models via the Ollama API to generate concise text summaries for efficient scene understanding and potential law enforcement use.

Detection and Mitigation of DDoS Attacks in an SDN-IoT Environment

- Developed a DDoS detection and mitigation framework for IoT networks using Software-Defined Networking (SDN), integrating Hop Count Filtering (HCF) to detect IP spoofing and counter-based methods to detect SYN flood based attacks.
- Implemented and tested the framework using Mininet, improving attack detection accuracy and enhancing IoT network security against DDoS attacks while minimizing false positives.

Compiler in C

- Implemented a compiler in C for a simple procedural programming language using LEX and YACC to evaluate basic expressions and arithmetic.
- Dealt with lexical, syntax and semantic analysis to produce intermediate code and machine code. Made use of symbol tables and activation records as well.

InstaHire

- Built a job board web app using Django & React, following Agile Scrum principles and employing the MVC architecture.

ToneSense

- Utilized Roberta Base machine learning models to analyze speech, accurately identifying and determining the primary emotions expressed in the content.

Flood Detection Project - Volunteer for a City Administration

- Built a web-based application and deployed it to collect information on rain levels for flood detection and alert for appropriate emergency-related actions.